

DAILY DIARY

Day 1 – Introduction and Setup

Attended orientation and learned about project objectives: combining **cryptography (data security)** and **steganography (data hiding)**. Installed Python and necessary libraries.

Code Concept: Print a welcome message to confirm setup.

```
print("SecureStegoPro Project - Setup Successful!")
```

Day 2 – Basics of Cryptography

Studied how encryption works to protect sensitive data. Learned about symmetric encryption using a key.

Code Concept: Encrypt and decrypt a simple text message.

```
from cryptography.fernet import Fernet
```

```
key = Fernet.generate_key()
```

```
cipher = Fernet(key)
```

```
token = cipher.encrypt(b"Hello World")
```

```
print(cipher.decrypt(token))
```

Day 3 – Understanding Steganography

Learned that steganography hides data inside images, audio, or text. Focused on pixel-based image hiding.

Code Concept: Convert text into binary for embedding.

```
text = "Hi"

binary = ''.join(format(ord(c), '08b') for c in text)

print(binary)
```

Day 4 – Reading and Writing Image Pixels

Understood how digital images store pixel values (RGB). Learned to read image data using Pillow.

Code Concept: Display pixel values of an image.

```
from PIL import Image

img = Image.open("cover.png")

print(img.getpixel((0,0))) # Shows RGB of pixel (0,0)
```

Day 5 – Least Significant Bit (LSB) Technique

Studied how to hide bits in the least significant part of an image pixel.

Code Concept: Change last bit of a pixel's red channel.

```
r = 100 # Example pixel red value

bit = 1

r = (r & ~1) | bit

print(r) # 101 means data embedded
```

Day 6 – Simple Data Embedding

Implemented a basic function to embed a short message into pixels.

Code Concept: Embed text bits into pixel list.

```
data = "A"

bits = ".join(format(ord(i),'08b') for i in data)

pixels = [(100,150,200)]

pixels[0] = (pixels[0][0] & ~1 | int(bits[0]),150,200)

print(pixels[0])
```

Day 7 – Simple Data Extraction

Learned how to extract hidden data by reading the least significant bit from pixels.

Code Concept:

```
r = 101 # example pixel value
```

```
hidden_bit = r & 1
```

```
print(hidden_bit)
```

Day 8 – Combining Encryption and Steganography

Planned integration: first encrypt data → then embed in image → extract → decrypt.

Code Concept: Show workflow idea.

```
def secure_stego(data):
```

```
    enc = cipher.encrypt(data)
```

```
    # embed enc in image
```

Day 9 – Encrypting Files

Encrypted text files using Fernet key to understand file-level encryption.

Code Concept:

```
with open("msg.txt","rb") as f:
```

```
    data = f.read()
```

```
enc = cipher.encrypt(data)

open("enc_msg.txt", "wb").write(enc)
```

Day 10 – Testing with Small Data

Encrypted small text data and embedded it into an image successfully.

Code Concept:

```
text = "Secret"

enc = cipher.encrypt(text.encode())

print(enc)
```

Day 11 – Understanding AES vs Fernet

Compared AES and Fernet encryption. Understood Fernet uses AES internally with simpler syntax.

Code Concept:

```
print("Fernet uses AES-128 + HMAC for message security.")
```

Day 12 – Writing Modular Code

Divided project into functions: `encrypt()`, `embed()`, `extract()`, `decrypt()`.

Code Concept:

```
def encrypt(msg,key): return Fernet(key).encrypt(msg.encode())
```

Day 13 – Integration of Modules

Integrated encryption + embedding. Verified complete workflow.

Code Concept:

```
key = Fernet.generate_key()
```

```
msg = "Karanveer"
```

```
enc = encrypt(msg,key)
```

```
print(enc)
```

Day 14 – Image Format Testing

Tested with PNG, BMP, and JPG. Found PNG best for lossless quality.

Code Concept:

```
formats = ["png","jpg","bmp"]
```

```
print("Best format:", formats[0])
```

Day 15 – Measuring PSNR (Quality)

Learned how to calculate image quality difference.

Code Concept:

```
import math

mse = 1

psnr = 20 * math.log10(255 / math.sqrt(mse))

print("PSNR:", psnr)
```

Day 16 – GUI Introduction

Studied basics of GUI using Tkinter for making SecureStegoPro user-friendly.

Code Concept:

```
import tkinter as tk

tk.Tk().title("SecureStegoPro")
```

Day 17 – Exception Handling

Learned to handle runtime errors using try-except.

Code Concept:

try:

```
    open("missing.txt")
```

except:

```
    print("File not found.")
```

Day 18 – Adding Checksum Validation

Added checksum to confirm data integrity after extraction.

Code Concept:

```
import hashlib
```

```
data = b"test"
```

```
print(hashlib.md5(data).hexdigest())
```

Day 19 – Testing with Multiple Inputs

Tested project using multiple text files and images for stability.

Code Concept:

```
for i in range(3):
```

```
    print("Test case",i+1,"passed")
```

Day 20 – Project Documentation Start

Started writing project report: *Introduction* and *Objectives*.

Code Concept:

```
sections = ["Intro","Objective"]  
  
for s in sections: print("Completed:", s)
```

Day 21 – Flowcharts and Architecture

Created diagrams for data flow and module connections.

Code Concept:

```
print("Flowchart created successfully")
```

Day 22 – Performance Testing

Performed speed and accuracy tests using time library.

Code Concept:

```
import time  
  
start = time.time()
```

```
# embedding here
```

```
print("Time:", time.time()-start)
```

Day 23 – Result Analysis

Summarized test results and checked PSNR, MSE, and recovery accuracy.

Code Concept:

```
results = {"PSNR":52.4,"Accuracy":"100%"}
```

```
print(results)
```

Day 24 – Code Review

Reviewed all files and cleaned unnecessary lines.

Code Concept:

```
print("Code cleaned and finalized")
```

Day 25 – Report Preparation

Completed report chapters and included screenshots of output.

Code Concept:

```
print("Report chapters merged successfully")
```

Day 26 – Writing Conclusion

Wrote *Conclusion and Future Scope*.

Code Concept:

```
future = ["GUI", "AI Integration", "Cloud Storage"]
```

```
print(future)
```

Day 27 – Documentation Review

Final proofreading of entire report for submission.

Code Concept:

```
print("Documentation reviewed successfully")
```

Day 28 – Presentation Work

Prepared PowerPoint slides for project evaluation.

Code Concept:

```
slides = ["Intro", "Design", "Results", "Future"]
```

Day 29 – Feedback Implementation

Made corrections suggested by mentor after review.

Code Concept:

```
print("Mentor feedback applied successfully.")
```

Day 30 – Final Submission and Presentation

Submitted final project and report. Mentor appreciated the practical implementation and neat documentation.

Code Concept:

```
print("SecureStegoPro completed and submitted  ")
```